Rajalakshmi Engineering College
Thandalam, Chennai.

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

DATA SCIENCE

AD23421 – INTERNSHIP

Internship Report submitted by

REGISTRATION NUMBER	**:**	**241801318**
STUDENT NAME	**:**	**VISHRUTHI K S**
YEAR / SEMESTER	**:**	**II / IV**
INTERNSHIP ORGANIZATION / COMPANY	**:**	**Altruisty Innovation Private Limited**
ONLINE / OFFLINE	**:**	**ONLINE**
TRAINER NAME	**:**	**PAMAJULA PRAVALIKA**
INTERNSHIP PERIOD	**:**	**FROM : 15/12/2025 TO: 31/12/2025**
INTERNSHIP DURATION	**:**	**15 DAYS (2 WEEKS)**

(Approved / Not Approved) **FACULTY ADVISOR**	(Approved / Not Approved) **EXAMINER 1**	(Approved / Not Approved) **EXAMINER 2**	(Approved / Not Approved) **HoD / AIDS**

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CHAPTER 1: ABSTRACT/ EXECUTIVE SUMMARY

The Iris Flower Classification project aims to develop a machine learning model capable of accurately identifying iris flower species based on their physical measurements. The project utilizes the well-known Iris dataset, which contains features such as sepal length, sepal width, petal length, and petal width to classify flowers into three species: Iris Setosa, Iris Versicolor, and Iris Virginica. Initially, the dataset was analysed and preprocessed through steps such as data cleaning, normalization, and feature preparation to ensure high-quality input for model training. Exploratory Data Analysis (EDA) was performed using Python libraries including Pandas, NumPy, Matplotlib, and Seaborn to understand patterns and relationships within the data. The dataset was then divided into training and testing sets, and a supervised machine learning algorithm, Logistic Regression, was implemented using the Scikit-learn library to build the classification model. The model's performance was evaluated using accuracy and confusion matrix metrics to ensure reliable predictions. After achieving satisfactory performance, the trained model was saved using Pickle for deployment. To make the system accessible and interactive, the model was integrated into a Flask-based web application that allows users to input flower measurements and obtain real-time predictions of the iris species. This project demonstrates the practical implementation of machine learning techniques for solving classification problems and highlights the importance of data preprocessing, model evaluation, and deployment in the data science workflow.

CHAPTER 2: INTRODUCTION

2.1 Brief Overview of the Organization:

Altruisty Innovation Private Limited is a technology-focused organization that delivers innovative solutions in emerging domains such as Artificial Intelligence, Generative AI, software engineering, and digital transformation. The company aims to bridge the gap between academic learning and industry requirements by providing structured training programs, internships, and real-time project exposure. It develops AI-driven applications including automated content generation, intelligent data analysis, workflow automation, and smart software systems. Through project-based learning, technical mentorship, and the use of modern development frameworks and tools, the organization promotes experiential learning and analytical thinking. During the internship period, structured sessions, guided project work, and continuous technical supervision were provided to ensure a clear understanding of practical AI workflows and professional standards.

2.2 Purpose of the internship:

The purpose of this internship was to develop practical competence in Artificial Intelligence and Data Science by gaining structured exposure to real-time project environments.

- Apply theoretical knowledge of Data Science in real-time structured project tasks.
- Enhance technical skills in Artificial Intelligence model development and implementation.
- Develop knowledge of professional workflows and industry-standard development practices.
- Prepare for professional roles in Artificial Intelligence and Data Science with exposure to real-world scenarios.

CHAPTER 3: OBJECTIVES OF THE INTERNSHIP

The objective of this Data Science internship was to gain practical exposure to Generative AI and data-driven technologies by applying theoretical concepts to real-time projects, understanding model development processes, and strengthening industry-relevant technical competencies.

The main learning goals during this internship were:

- To develop a comprehensive understanding of Generative AI concepts, including large-scale model training, input processing, and contextual content generation.
- To enhance the ability to design clear, structured, and effective prompts for accurate AI-generated outputs.
- To apply theoretical knowledge by building small-scale AI-driven applications using appropriate programming tools and frameworks.
- To gain hands-on experience with diverse AI tools and evaluate their performance, capabilities, and practical use cases.
- To strengthen programming skills, analytical reasoning, and systematic problem-solving abilities through continuous experimentation and refinement.

CHAPTER 4: BRIEF DESCRIPTION OF WORK / PROJECT

Project Title:

"Iris Flower Classification using Logistic Regression"

4.1 Tasks Handled in the Internship:

- Collected and understood the Iris Dataset containing sepal length, sepal width, petal length, and petal width features.
- Performed data preprocessing including checking for missing values and validating feature distributions.
- Conducted exploratory data analysis to understand feature relationships and class separability.
- Split the dataset into training and testing sets.
- Implemented a Logistic Regression model for multi-class classification.
- Trained the model using training data and tested it on unseen data.
- Evaluated model performance using accuracy score and confusion matrix.
- Interpreted classification results and analyzed model efficiency.

4.2 Methods Used:

- Supervised Learning Approach – Since the dataset contains labeled outputs (Setosa, Versicolor, Virginica).
- Logistic Regression Algorithm – A statistical classification algorithm used to predict categorical outcomes using probability estimation.
- Data Splitting Technique – Train-test split method to validate model performance.
- Model Evaluation Metrics – Accuracy score and confusion matrix to measure classification performance.

4.3 Tools Used:

Programming Language:

- **Python** is used as the main programming language to develop the Iris flower classification model because it provides simple syntax and strong support for data science libraries.

Development Environment:

- **Visual Studio Code** is used for writing, executing, and testing the program in an efficient development environment, making it easier to manage code and view outputs during execution. NumPy is used for performing numerical computations and handling arrays required for mathematical operations in the model.
- **Git & GitHub** is a version control system used to track changes in the source code, while GitHub is a platform used to store and manage the project repository. In this project, they were used to manage code, track updates, and maintain version history during development.

Framework:

- **Flask** is a simple and lightweight Python web framework used for developing web applications. In this project, it helps integrate the machine learning model with the website and processes the inputs provided by the user.

Libraries:

- **Pandas** is used for loading, organizing, and preprocessing the dataset in a structured DataFrame format.
- **Matplotlib** and Seaborn are used for data visualization to plot graphs and understand feature relationships.
- **Scikitlearn** is used to implement the Logistic Regression algorithm and evaluate the model using accuracy and other performance metrics.
- **Joblib** is a Python library used to save and load trained machine learning models efficiently. In this project, it was used to store the trained model as a **.pkl** file and later load it into the Flask application for making predictions.

Frontend Technologies:

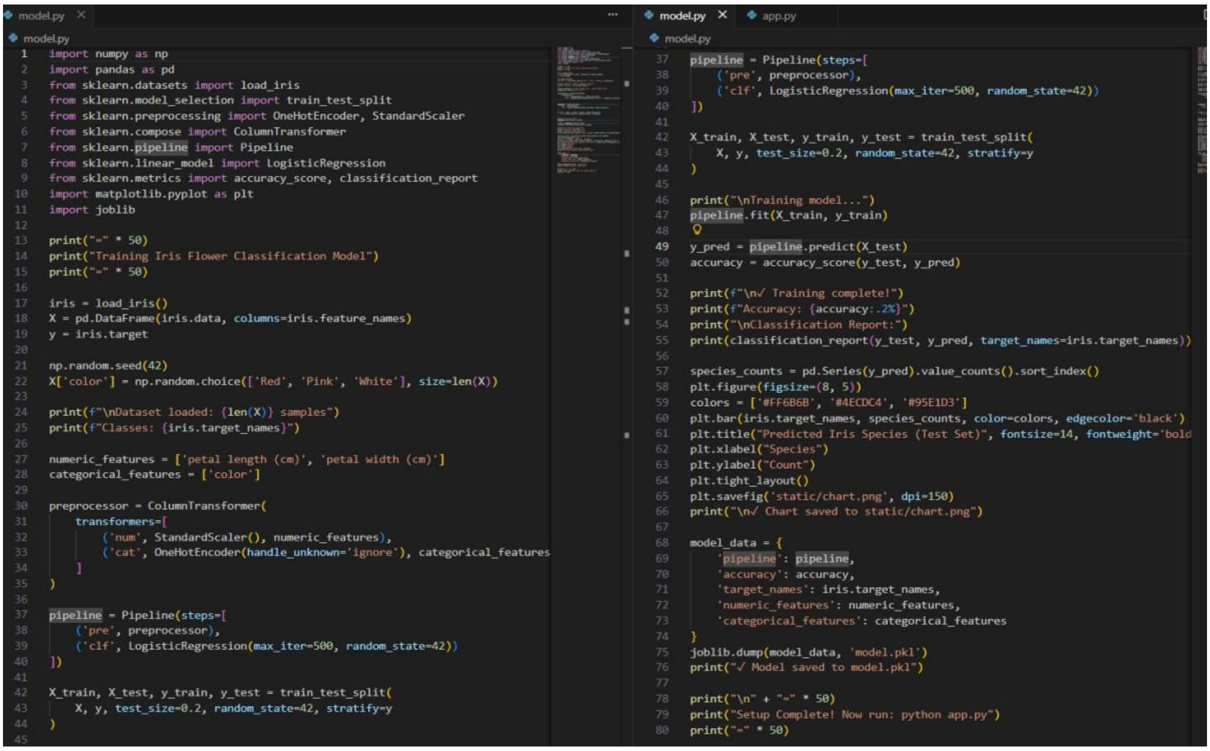
- **HTML** is used to structure the content of web pages. In this project, it was used to design the user interface for entering flower details and displaying the prediction results.
- **CSS** is used to style and format web pages. It was used to improve the appearance of the web application by adding layout, colors, fonts, and other visual elements.

4.4 Iris Flower Classification System Implementation:

The Iris Flower Classification system uses machine learning to identify the species of an Iris flower based on features like sepal length, sepal width, petal length, and petal width. Logistic Regression is used to train the model for accurate classification. A pipeline is applied to combine feature scaling and classification for better performance. The trained model is deployed using a Flask web application that allows users to enter flower measurements and get predictions.

4.4.1 Logistic Regression for Iris Classification:

The Iris Flower Classification using Logistic Regression was the project focuses on identifying the species of an Iris flower based on its features such as sepal length, sepal width, petal length, and petal width. It uses the Logistic Regression algorithm, a supervised machine learning technique, to classify the flowers into different categories.



model.py

```python
import numpy as np
import pandas as pd
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import OneHotEncoder, StandardScaler
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, classification_report
import matplotlib.pyplot as plt
import joblib

print("=" * 50)
print("Training Iris Flower Classification Model")
print("=" * 50)

iris = load_iris()
X = pd.DataFrame(iris.data, columns=iris.feature_names)
y = iris.target

np.random.seed(42)
X['color'] = np.random.choice(['Red', 'Pink', 'White'], size=len(X))

print(f"\nDataset loaded: {len(X)} samples")
print(f"Classes: {iris.target_names}")

numeric_features = ['petal length (cm)', 'petal width (cm)']
categorical_features = ['color']

preprocessor = ColumnTransformer(
    transformers=[
        ('num', StandardScaler(), numeric_features),
        ('cat', OneHotEncoder(handle_unknown='ignore'), categorical_features
    ]
)

pipeline = Pipeline(steps=[
    ('pre', preprocessor),
    ('clf', LogisticRegression(max_iter=500, random_state=42))
])

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)
```

model.py

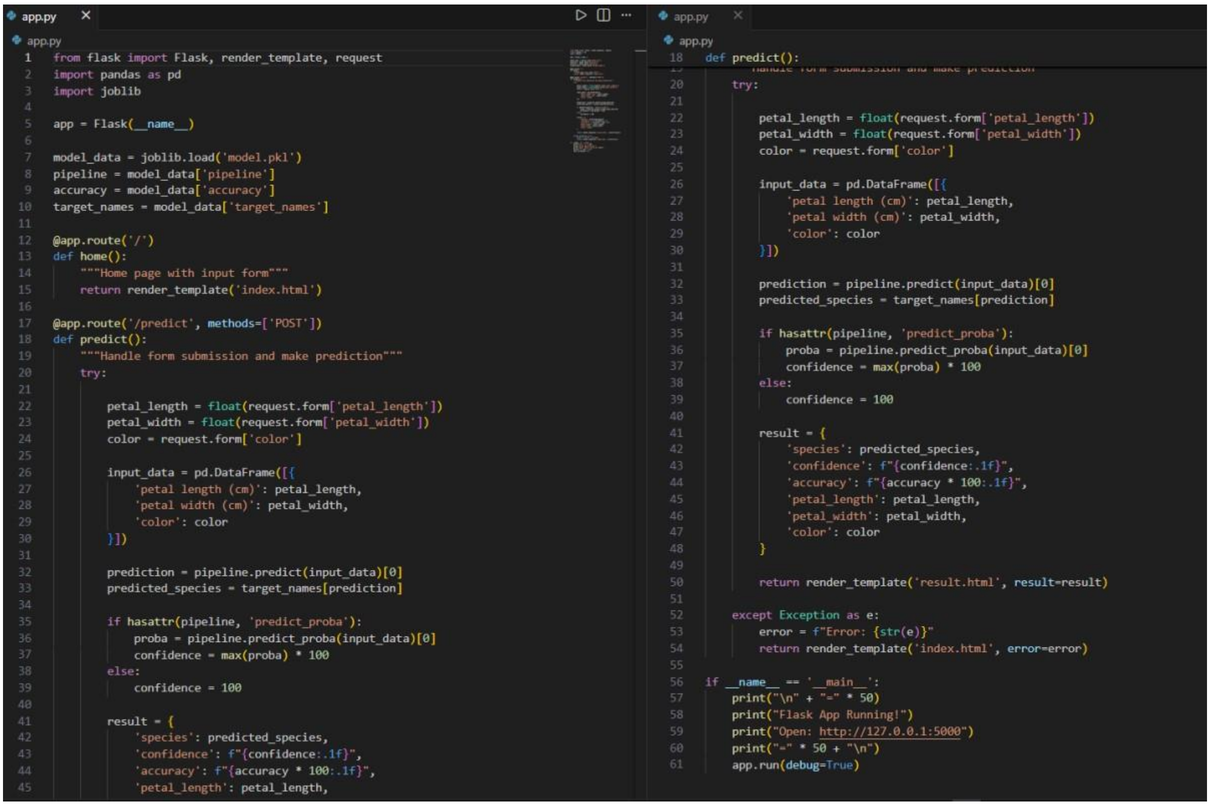
```python
pipeline = Pipeline(steps=[
    ('pre', preprocessor),
    ('clf', LogisticRegression(max_iter=500, random_state=42))
])

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)

print("\nTraining model...")
pipeline.fit(X_train, y_train)

y_pred = pipeline.predict(X_test)
accuracy = accuracy_score(y_test, y_pred)

print(f"\n✓ Training complete!")
print(f"Accuracy: {accuracy:.2%}")
print("\nClassification Report:")
print(classification_report(y_test, y_pred, target_names=iris.target_names))

species_counts = pd.Series(y_pred).value_counts().sort_index()
plt.figure(figsize=(8, 5))
colors = ['#FF6B6B', '#4ECDC4', '#95E1D3']
plt.bar(iris.target_names, species_counts, color=colors, edgecolor='black')
plt.title("Predicted Iris Species (Test Set)", fontsize=14, fontweight='bold
plt.xlabel("Species")
plt.ylabel("Count")
plt.tight_layout()
plt.savefig('static/chart.png', dpi=150)
print("\n✓ Chart saved to static/chart.png")

model_data = {
    'pipeline': pipeline,
    'accuracy': accuracy,
    'target_names': iris.target_names,
    'numeric_features': numeric_features,
    'categorical_features': categorical_features
}
joblib.dump(model_data, 'model.pkl')
print("✓ Model saved to model.pkl")

print("\n" + "=" * 50)
print("Setup Complete! Now run: python app.py")
print("=" * 50)
```

Fig 4.4.1: **Pipeline Implementation For Iris Classification**

Fig 4.4.1 shows the implementation of the machine learning model in model.py. It includes data preprocessing steps and the use of a Pipeline to combine feature scaling and the Logistic Regression algorithm. The figure represents how the model is trained and prepared for accurate Iris flower classification.

4.4.2 Flask Application for Iris Classification:

Flask is used to build a web-based application for predicting the species of an Iris flower. It shows that the project integrates a trained Logistic Regression model with a user-friendly interface. Users can enter flower measurements through the web page, and the application processes the input and displays the predicted Iris species as the result.



```python
from flask import Flask, render_template, request
import pandas as pd
import joblib

app = Flask(__name__)

model_data = joblib.load('model.pkl')
pipeline = model_data['pipeline']
accuracy = model_data['accuracy']
target_names = model_data['target_names']

@app.route('/')
def home():
    """Home page with input form"""
    return render_template('index.html')

@app.route('/predict', methods=['POST'])
def predict():
    """Handle form submission and make prediction"""
    try:

        petal_length = float(request.form['petal_length'])
        petal_width = float(request.form['petal_width'])
        color = request.form['color']

        input_data = pd.DataFrame([{
            'petal length (cm)': petal_length,
            'petal width (cm)': petal_width,
            'color': color
        }])

        prediction = pipeline.predict(input_data)[0]
        predicted_species = target_names[prediction]

        if hasattr(pipeline, 'predict_proba'):
            proba = pipeline.predict_proba(input_data)[0]
            confidence = max(proba) * 100
        else:
            confidence = 100

        result = {
            'species': predicted_species,
            'confidence': f"{confidence:.1f}",
            'accuracy': f"{accuracy * 100:.1f}",
            'petal_length': petal_length,
            'petal_width': petal_width,
            'color': color
        }

        return render_template('result.html', result=result)

    except Exception as e:
        error = f"Error: {str(e)}"
        return render_template('index.html', error=error)

if __name__ == '__main__':
    print("\n" + "=" * 50)
    print("Flask App Running!")
    print("Open: http://127.0.0.1:5000")
    print("=" * 50 + "\n")
    app.run(debug=True)
```

Fig 4.4.2: Python Flask Application Code for Iris Flower Prediction

Fig 4.4.2 shows the Flask-based web application used to deploy the trained Iris Flower Classification model. The saved model is loaded using Joblib and configured to accept user inputs. Feature values are processed and passed to the trained pipeline for prediction. The application displays the predicted species along with model performance details in a real-time web interface.

4.4.3 Web Interface Design:

The user interface of the Iris Flower Classification system is developed using HTML, CSS, and JavaScript. It provides a web form where users can enter flower measurements and select the color. CSS is used for styling and responsive design, while JavaScript adds interactive features like sliders and color selection. The input data is sent to the Flask backend to predict and display the Iris flower species.

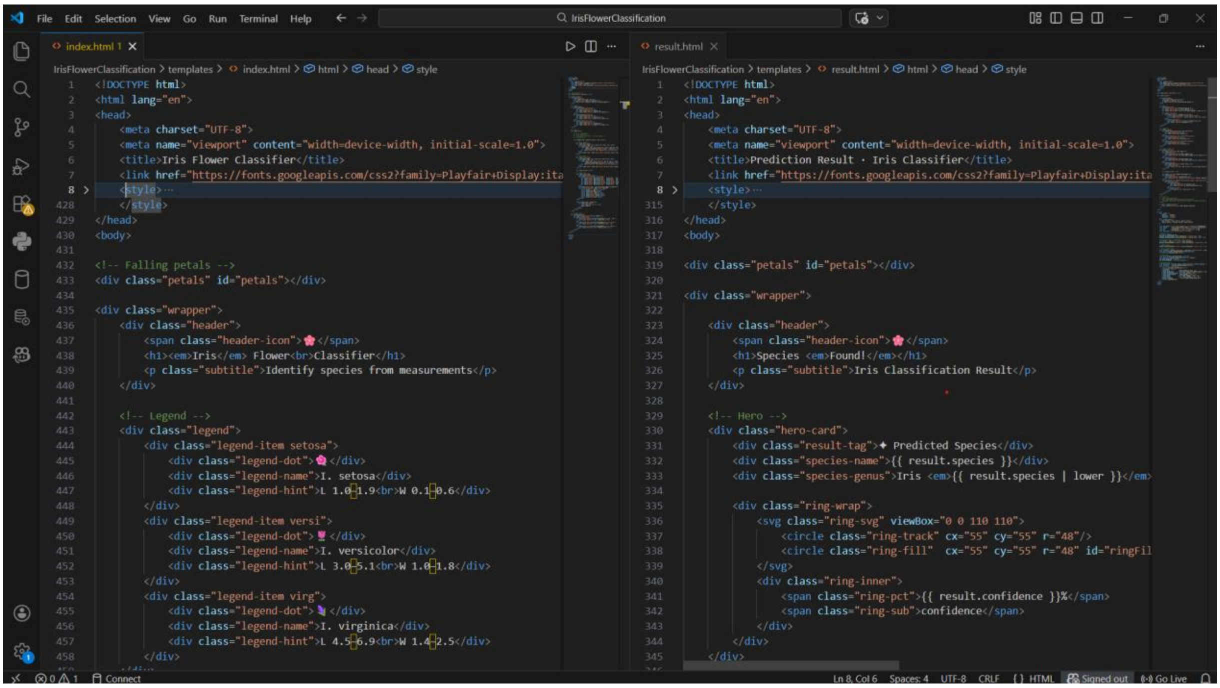


Fig 4.4.3: HTML Template Implementation in Visual Studio Code

Fig 4.4.3 shows the implementation of the HTML templates in Visual Studio Code used for the Iris Flower Classification system. The templates include the input page (index.html) and the result page (result.html), which provide the user interface for the web application. These templates allow users to enter flower measurements and display the predicted Iris species along with related details through the Flask web application

4.5 RESULT:

During the internship, the project was successfully completed by developing and deploying the Iris flower classification model through a Flask web application. The system was tested and the final output was generated successfully through the web interface.

- User interface is displayed to enter flower measurements
- Prediction result is shown on the webpage
- Logistic Regression model integrated with the web app
- Model accuracy graph to show performance and accuracy

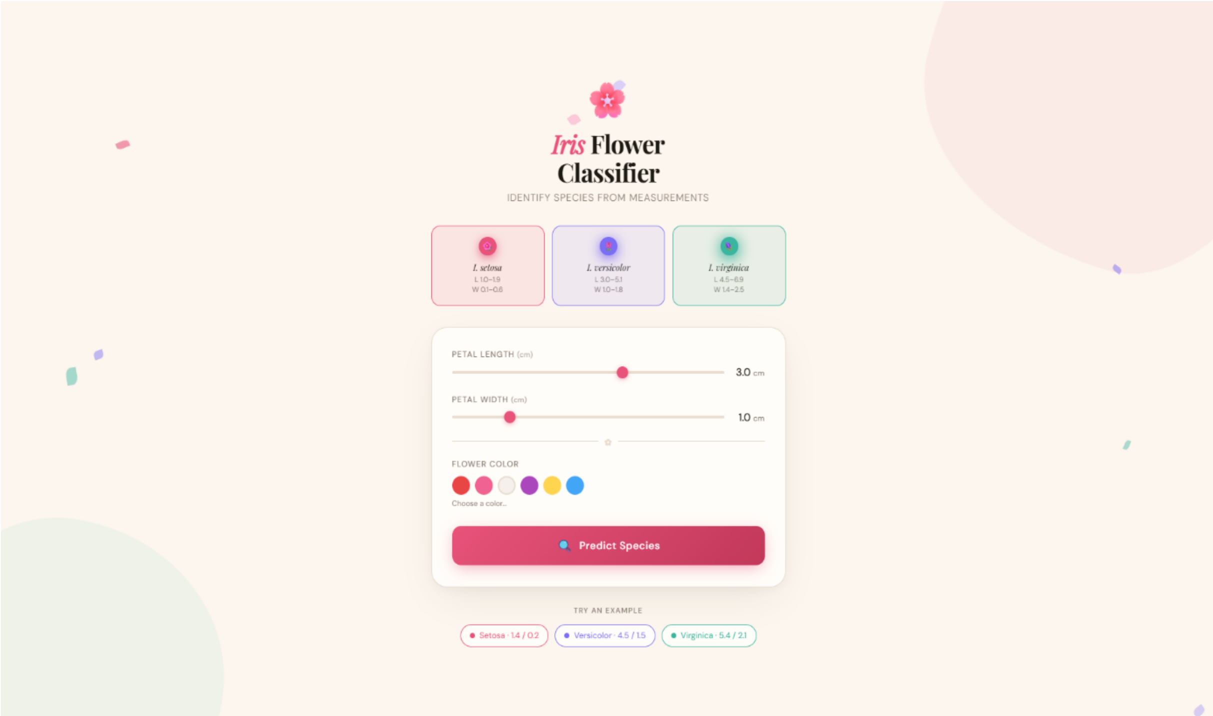


Figure 4.5.1: Input Interface of Iris Flower Classification System

Figure 4.5.1 The web interface of the Iris Flower Classification system provides a simple and interactive platform for users to input flower features and obtain predictions. Users can enter petal length and petal width using slider controls and select the flower colour through colour options. The interface is designed to be visually appealing, responsive, and easy to use. After submitting the inputs, the system processes the data and displays the predicted Iris species along with related prediction details on the result page.

Additionally, example input options are provided to help users quickly test the system with sample values. The overall design improves user experience and enables efficient interaction with the classification system.

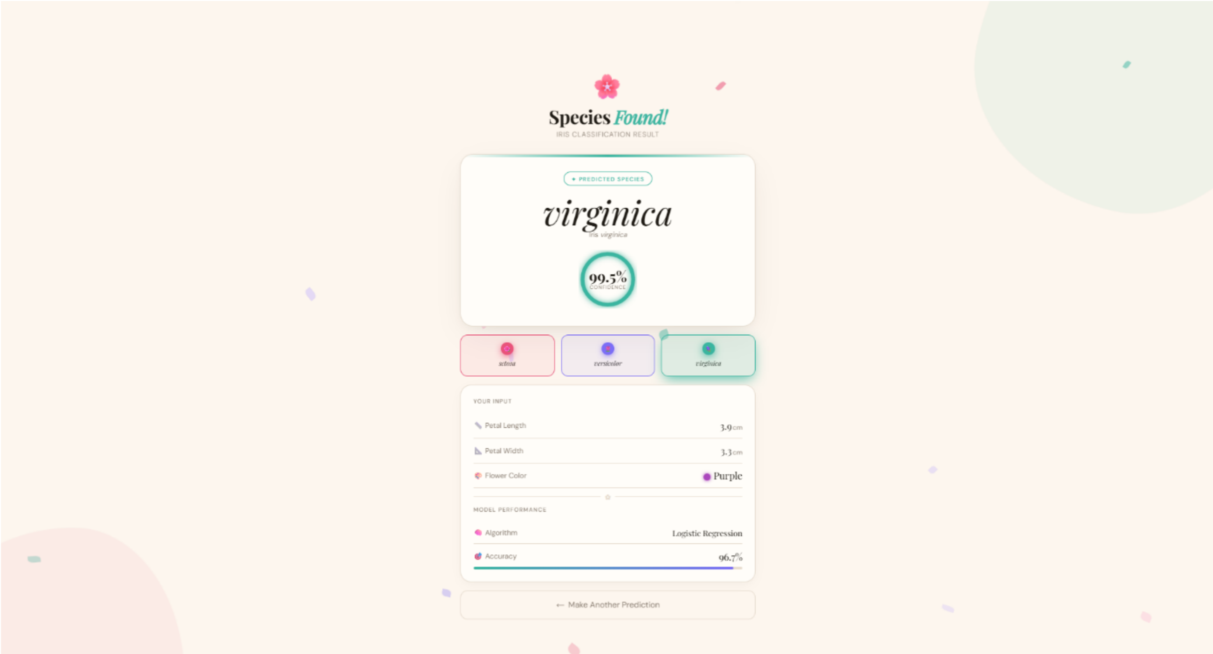


Figure 4.5.2: Iris Flower Classification Prediction Result Interface

Figure 4.5.2 demonstrates the prediction result interface of the deployed Iris Flower Classification system. The interface clearly displays the predicted Iris species along with the confidence score generated by the machine learning model. It also presents the user input details such as petal length, petal width, and selected flower colour, allowing users to verify the values used for prediction. In addition, the overall accuracy of the trained model is shown to indicate the performance of the classification system.

The result page is designed to present the prediction in a visually organized and easy-to-understand format. A graphical indicator is used to highlight the confidence level of the predicted species. The interface also provides an option to make another prediction, enabling users to quickly return to the input page and test the system with different values. This result interface demonstrates the successful integration of the trained machine learning model within a web-based application for real-time Iris flower species classification.

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CHAPTER 5: LEARNING OUTCOME

The successful completion of this project has resulted in the following technical proficiencies and key competencies developed:

- Understanding the fundamental concepts of Machine Learning and their application in classification problems using the Iris dataset.
- Interpreting the Iris dataset and explaining how features like sepal length, sepal width, petal length, and petal width are used to classify flower species.
- Applying data preprocessing techniques such as data cleaning, encoding, and feature scaling to improve model accuracy.
- Training and evaluating classification algorithms like Logistic Regression for predicting Iris flower species (Setosa, Versicolor, Virginica).
- Integrating machine learning models into Flask-based web applications for real-time flower classification.
- Strengthening logical thinking and problem-solving skills through the implementation of an end-to-end classification system.
- Handling and managing structured data for storing input features and prediction results effectively.
- Developing confidence in building and deploying machine learning-based applications for real-world classification problems

CHAPTER 6: CONCLUSION

The Web-Based Iris Flower Classification System successfully demonstrates the practical application of Machine Learning techniques in solving classification problems. By utilizing a Logistic Regression model, the system is capable of analyzing flower features such as sepal length, sepal width, petal length, and petal width, and providing accurate species predictions in real time through a Flask-based web application. The project highlights the importance of data preprocessing, model training, evaluation, and deployment in building efficient classification systems. The integration of structured data handling ensures proper management of input features and prediction results, making the system reliable and scalable. Overall, the project provided valuable hands-on experience in developing an end-to-end machine learning solution, strengthening technical knowledge in Python, data analysis, web development, and model integration while demonstrating how machine learning can be effectively used for pattern recognition and prediction tasks.

CHAPTER 7: INTERNSHIP COMPLETION CERTIFICATE

ALTRUISTY
Innovation Pvt Ltd

CERTIFICATE OF COMPLETION

DATE: 07-01-26

This is to Certify that Vishruthi K S has Successfully Completed a 15 Days of Internship at Altruisty Innovation Pvt Ltd, from 15-12-2025 to 31-12-2025, in the Domain of Data Science .

Throughout the Duration of the Internship, Vishruthi K S has Demonstrated Remarkable Growth and Development, Gaining Valuable Experience and Insights Into The Field of Data Science .

Their commitment to learning and adapting to new challenges reflects Altruisty's core values of excellence and innovation.

We hereby acknowledge Vishruthi K S for This outstanding performance and dedication during the internship tenure.

Best Regards,

A.N. Bhavithra

Bhavithra A N
HR & Executive Lead

ALTRUISTY

REG:1252025598

8610604326 altruistyinnovation.com
altruistybusiness@gmail.com

8. INTERNSHIP FEEDBACK